

CUSTOMER MILESTONE BRIEF

MS #22 - Mixed PL/I Modernization Proof

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.22.0
Date	2026-08-26
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #22 - Mixed PL/I Modernization Proof converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Demonstrates a bounded PL/I, COBOL, and DB2 modernization slice with explicit cross-language contracts.

Why this matters to a customer

Demonstrates a bounded PL/I, COBOL, and DB2 modernization slice with explicit cross-language contracts.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Advanced the bounded mixed ACCTPL1 PL/I-COBOL-Db2 workload from discovery-only to a content-addressed local development proof.
- Made the PL/I source explicit about OPTIONS(COBOL), the CBACT04C parameter aggregate, and DIVIDE(...,5,2) truncation semantics.
- Added a curated behavior contract, seven boundary fixtures, a source-faithful executable oracle, and an independently implemented Python modernization candidate.
- Added nine mutation probes covering Db2 overwrite behavior, risk calculation and rounding, COBOL target and parameters, and fail-closed error side effects.
- Added a production-shaped Java service and JUnit suite for the same bounded service seam.
- Added deterministic POSIX and PowerShell build/verify launchers plus a fail-closed development receipt binding graph, fragment, contract, fixtures, comparison, and candidate sources.
- Promoted PL/I capability gates 3-5 to passed and gate 7 to mechanism-ready while keeping authorized z/OS execution, signed equivalence, mainframe equivalence, and production readiness explicitly blocked.

How it differs from earlier work

MS #22 builds on MS #21 (Mainframe Access Readiness Campaign). The earlier milestone established its own bounded capability; this milestone adds mixed pl/i modernization proof while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use mixed pl/i modernization proof as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.22.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity